JAMES POIRIER

WORK EXPERIENCE

PRINCIPAL AI ENGINEER
Mar 2024-Present

RTX - CODE Center

- Engineering lead implementing on-prem and cloud-native AI and cybersecurity solutions
- Designed, implemented and deployed RAG and Agent-based applications using Ollama, AutoGPT, LangChain, and Kubernetes to modernize cyber solutions and trainings
- Designing algorithms and reporting methods to predict future FOSS vulnerability risks across project dependency trees and detect injected vulnerabilities in compiled binaries
- Responsible for $3M worth of cyber/AI Internal Research & Development (IRAD)
- Carnegie Mellon and UC Irvine Capstone coordinator and Cyber Ops and Dev (CODE) Center University Relations representative, leading 1-2 capstone projects annually

SENIOR ML ENGINEER
May 2022 - Mar 2024

Raytheon (Acquired by RTX)

- Led development of an AI-driven data filtration platform for cybersecurity analysts, leveraging Python, PyTorch, and NLP techniques to increase threat detection by 30%
  - Engineered data ingestion pipelines integrating Neo4j, MongoDB, and ChromaDB for vectorized storage and graphical correlation revealing insights to threat patterns
- Architected eVTOL Advanced Air Mobility system in consultancy with Hyundai Supernal
  - Developed FAA compliant takeoff, path planning and object detection systems
- Authored Basis of Estimates (BOEs) and submitted 14 successful project proposals

MACHINE LEARNING ENGINEER
Feb 2021 - May 2022

Coherent Technical Services Inc.

- Developed object detection, semantic segmentation, path planning, and sensor fusion algorithms in Python and C++ for autonomous trains with Union Pacific and BNSF
- Created a Python-based data collection and annotation toolkit using OpenCV, YOLO models, numpy, and scikit-image accelerating data annotation processes by 600%
- Responsible for ingesting, processing, and distributing over 20TB of vehicle data daily

SOFTWARE ENGINEER
Jan 2019 - Nov 2020

California Cybersecurity Institute

- Co-Created the California Cyber Innovation Challenge for 4000 competitors
- Developed a scalable competition-hosting app using Python, React and AWS EC2
- DEFCON 2019, 2020 Aerospace Village and Bio-Hacking Village CTF Designer

DATA ENGINEER
June 2019 - Sept 2019

Dragos, Global Emancipation Network (GEN)

- Implemented web scrapers, data processing/ingestion pipelines, and RNNs in Python
- Collaboratively worked to create a first-of-its-kind investigative analytics platform to locate potential victims of human trafficking through the dark web
- Identify 100,000 victims per year using government furnished resources

RESEARCH AND PUBLICATIONS

ENHANCING KNOWLEDGE TRANSFER WITH AI
HICSS: 2024

- Lead author - Exploration into RAG applications within Education & Industry

ANN TO DETECT ALZHEIMERS IN MRI SCANS
Cal Poly: 2020

- Detecting Alzheimer's development signs with 88% accuracy using InceptionV3

CERTIFICATIONS

Machine Learning Specialization	Stanford: 2024
Deep Learning Specialization	Stanford: In Progress
Hands-On-Hacking	Blackhat: 2023
AWS Technical Professional	AWS: 2020
DoD Secret Clearance	2021
Eagle Scout	2014

PERSONAL INFO

San Jose, CA

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EDUCATION

B.S. Electrical Engineering

California Polytechnic State University, San Luis Obispo

Minor: Computer Science

June 2020

SOFTWARE SKILLS

Languages:

Python	C++	SQL
C	Java	Rust

Frameworks:

PyTorch	TensorFlow
ChromaDB	ONNX
Flask	Node.js
LangChain	AutoGPT

DevOps Skills:

Git	Docker	AWS
Jira	Kubernetes	Jenkins

TALKS & PANELS

HICSS 56, TAPIA 2024, AI-ISAC 2024, REDx 2023, DEFCON 27 2019

INNOVATIONS

Decision Engine for Software Integrity and Releasability

Filed Sept 2024, Patent Pending.

Software Releasability Workflow

Filed April 3, 2025

Application #20250110730

6 Invention Disclosures

Awarded through RTX

OTHER WORK

Course Tester | DeepLearning.ai

June 2024-Present

CS and ICS Mentor | UC Irvine

Nov 2023-Present

CS Lecturer | Outpour Movement

June-Sept 2018